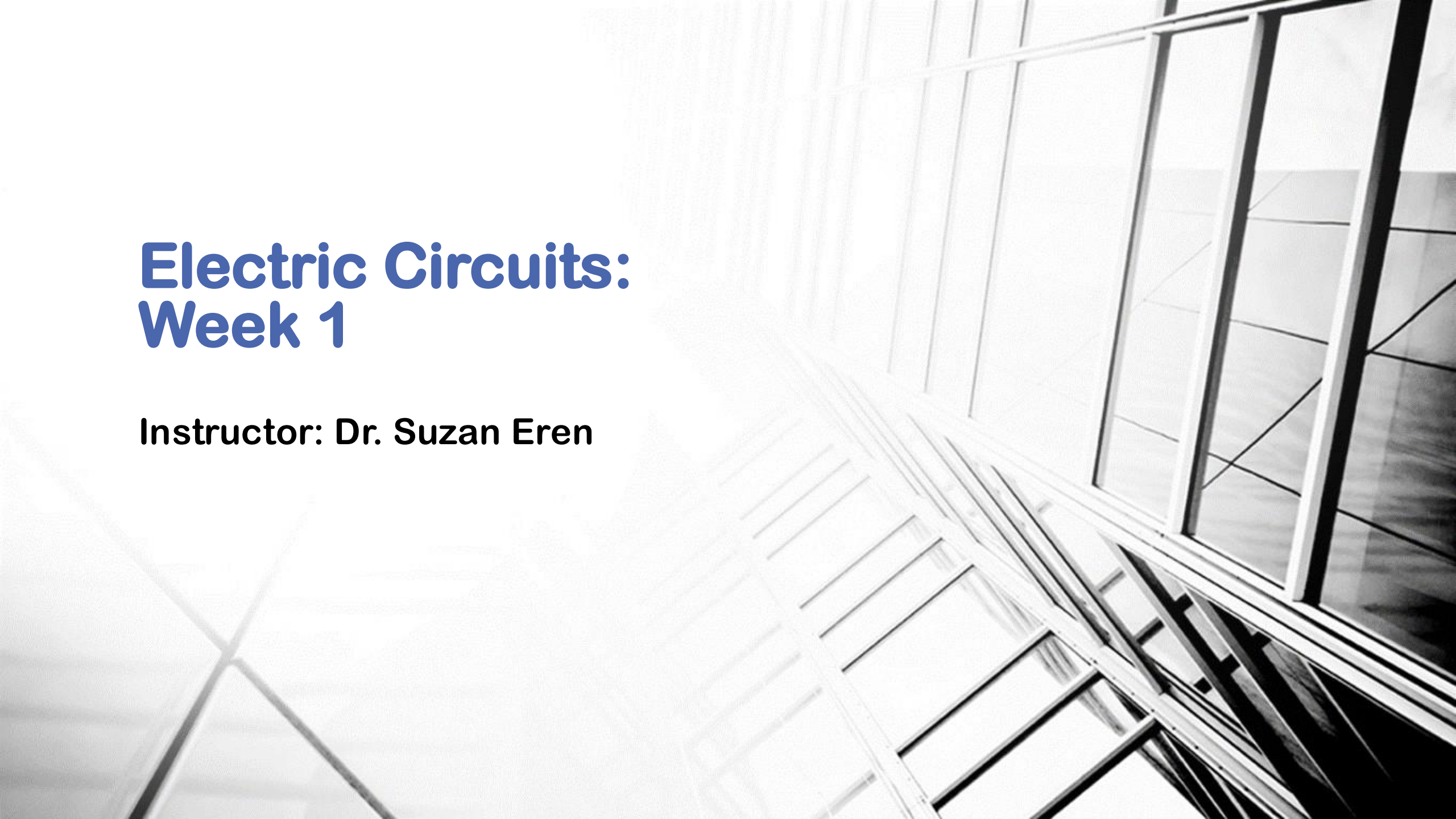
Electric Circuits: Week 1

Instructor: Dr. Suzan Eren



Welcome to Electric Circuits!

Email: suzan.eren@queensu.ca

Office Hours: **Thursdays** from 10:30am-11:30am @ WLH Rm 406

TA Contact Information:

Name: Prince Keelson
Email: 24cbtt@queensu.ca
Office Hours: Wednesdays from 4:00pm-5:00pm @WLH 110

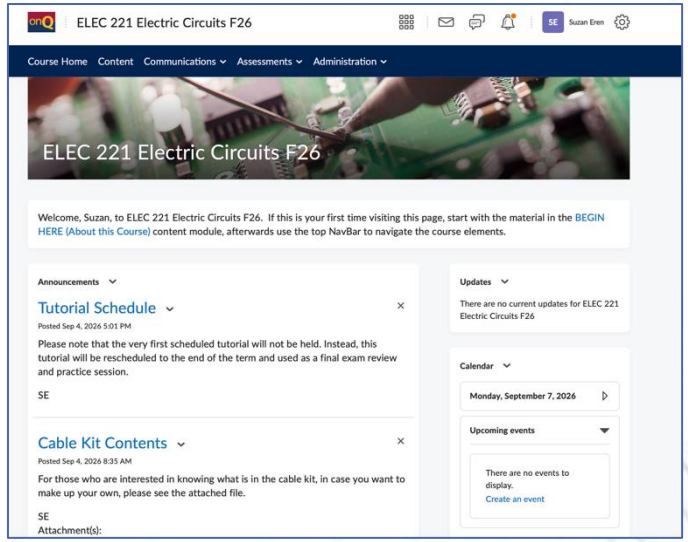
Name: Prosper Allo
Email: 22xmg@queensu.ca
Office Hours: Mondays from 11:00am-12:00pm@TBD

Question: Any volunteers for **Class Representative**?

Introduction: Course Website

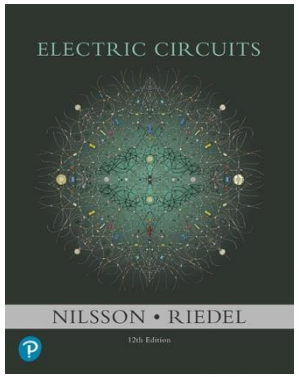
OnQ: The course website is live and will contain the following information:

- ***Announcements → check regularly and sign up for email notifications!***
- Syllabus (check the "Begin Here" tab)
- Course Slides
- Tutorials
- Labs



Introduction: Textbook

Textbook: *Electric Circuits*
By: Nilsson and Riedel
12th Edition



NOTE: There are three hard copies of the 11th-edition on reserve @ Stauffer Library for 3-hour term loan.

Material Covered: Chapters 1-4, 6-10 and Appendices: A, B, F, H

Textbook can be purchased as an eText from the Campus Bookstore website:

https://campusebookstore.com/link/?id=98a9c8cf-fa8c-4008-a610-ed6897d3b33c&ref=7586b56c-5a1b-4be0-bd7a-8d6cda96cdd0&sid=ze8d1w7xqk1hr3vxg9jva57a

eText cost is $80.99 CAD + Tax

Introduction: Tutorials and Grading

Tutorials: **Your second scheduled tutorial is going to be your first tutorial. An exam prep session will be added at the end of the term to make up for the missed first tutorial.**

Grading:	Labs	30%	(6 labs, 5% each)
	Midterms	25%	(2 midterms, 12.5% each)
	Final Exam	45%	

Midterms: Midterms will take place on the following dates. Check your OnQ announcements regularly for further information regarding location and time.

Midterm 1 – Friday, October 23rd
Midterm 2 – Friday, November 27th

Introduction: Labs

Here are a few important things to note regarding labs:

1. **Procure cable kits for the labs before attending your first lab. These can be bought from the Campus Bookstore, or bought/borrowed from upper year students.**

2. **There are six lab sections (sections 005-010). Consult with the SOLUS lab schedule on OnQ to determine when labs begin for your lab section.**

3. Those in **Lab Section 005** have a lab scheduled on Labour Day. Since this is a holiday, it has been rebooked for Wednesday, September 16th from 6:30-9:30pm in BMH 314. Other lab dates remain unchanged.

Introduction: Labs

4. **Lab instructions will be posted on OnQ.**
5. **Complete the pre-lab before attending the lab. They will be graded for completion.**
6. **Lab grades will be determined by the following:**
    1. **Pre-Lab Report (20%): Pre-labs must be finished individually before the lab takes place and will be checked for completion during the lab. They must also be submitted to OnQ by the due date along with the team post-lab report.**
    2. **Post-Lab Report (60%): One post-lab report is required per team. Post-lab reports should be submitted to OnQ by the due date, which is one week after your session.**
    3. **Attendance (20%): Attendance is taken at the start of each lab and is mandatory to receive credit for the lab. There is a 50% reduction in your attendance grade if you arrive more than 15 minutes late to your lab. Late by >60 minutes: 0% attendance.**

Safety

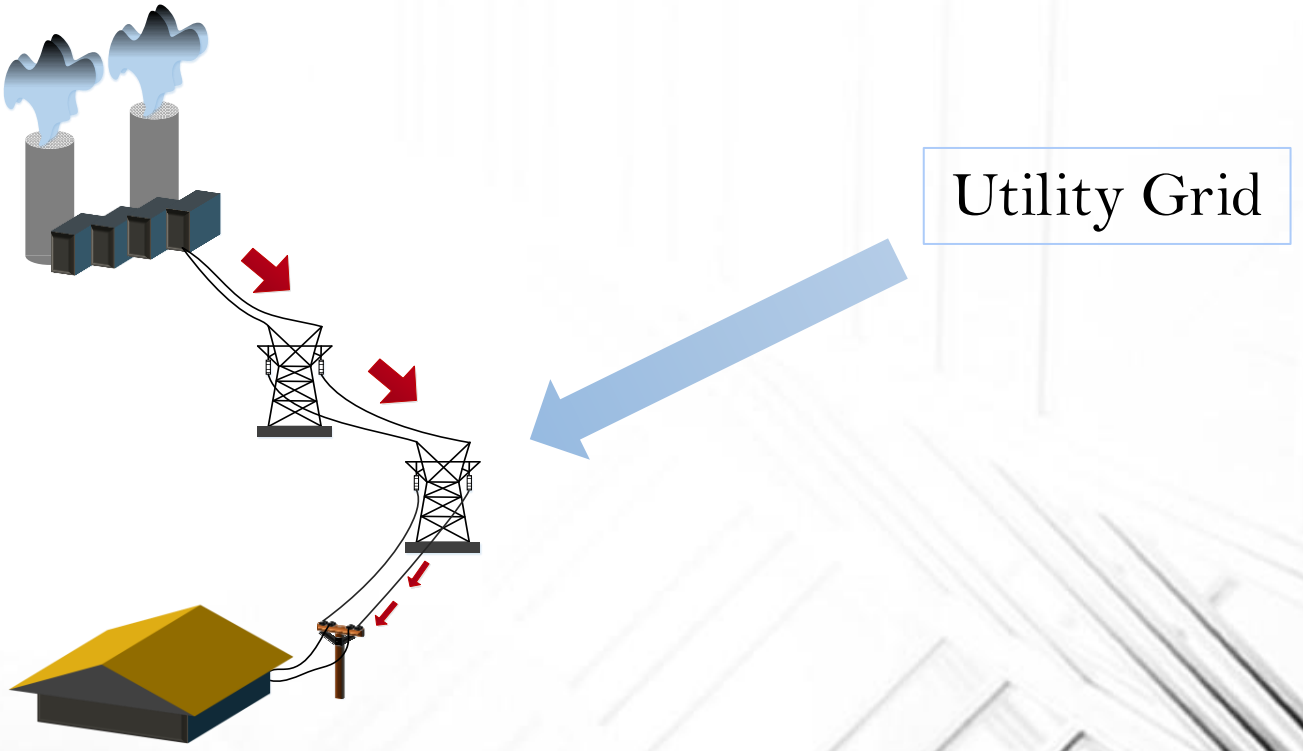
- Please take note of your nearest exits
- If you have any safety concerns, please don't hesitate to bring it to my attention and/or a TA's attention
- In case of an emergency please call 911
- Campus Emergency Response can be reached by dialing (613) 533-6111 (add to your phone contacts) on your phone, or by dialing 36111 on any internal CAMPUS phone

Introduction

What? A basic electric circuit is a closed loop with a ***electric power source***, a ***load***, and a ***conductive path*** for flowing electrons.

Why? Electric circuits are used in **many** applications. Here are some examples…

Power Systems



Introduction

Communication Systems: Cell phones, radio, radars



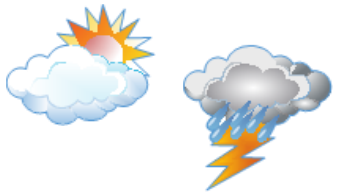
Computer Systems: Laptops, desktops, tablets



Control Systems: Airplanes, drones, robotics



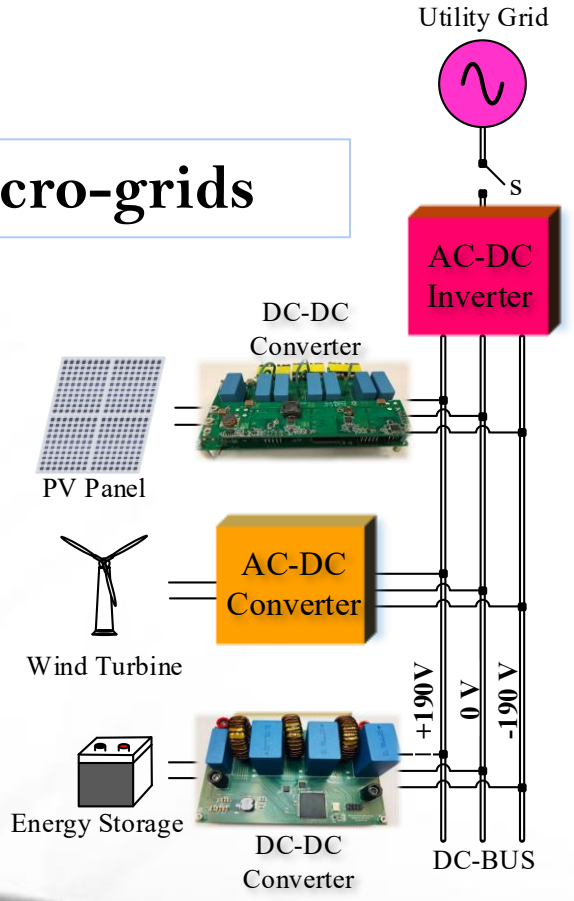
Signal Processing Systems: Ultrasounds, MRI, CT scans, weather forecasting



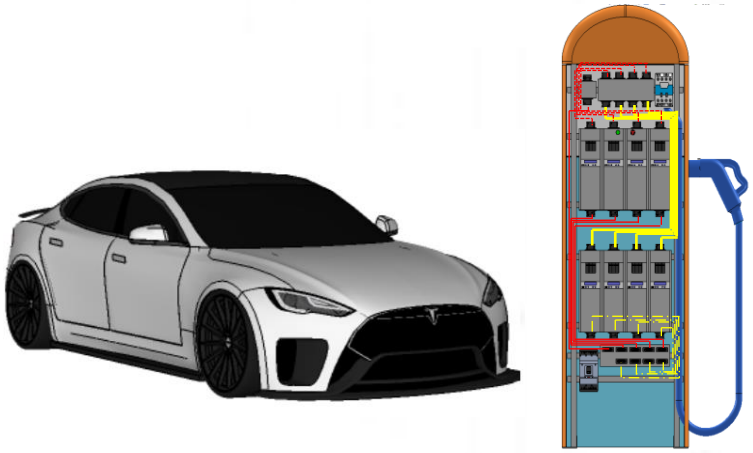
Introduction

Here are a few applications I have worked on in the past…

Micro-grids



Electric Vehicles



E-Brakes for Locomotives

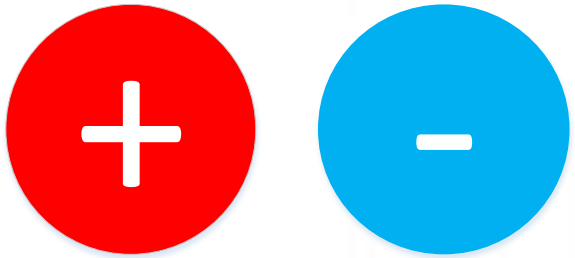


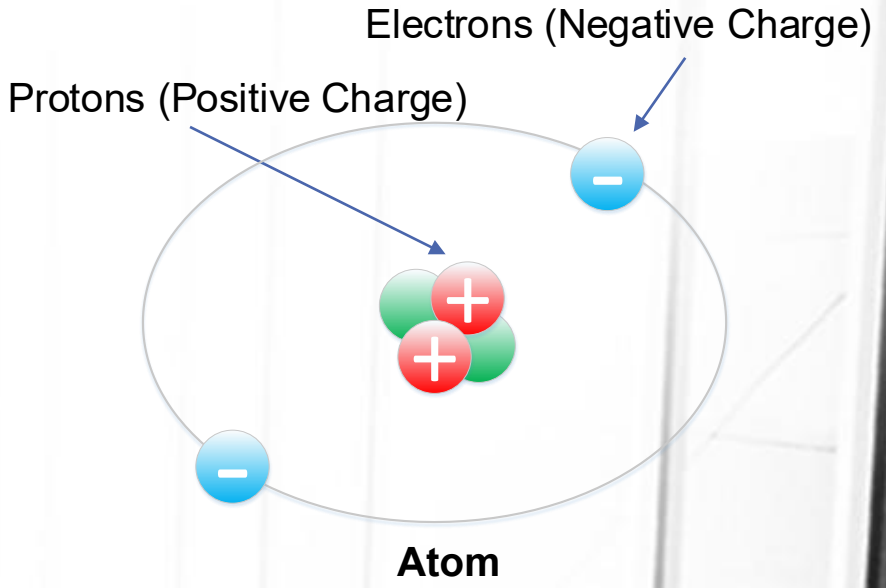
Kinetic Energy=$0.5mv^2$



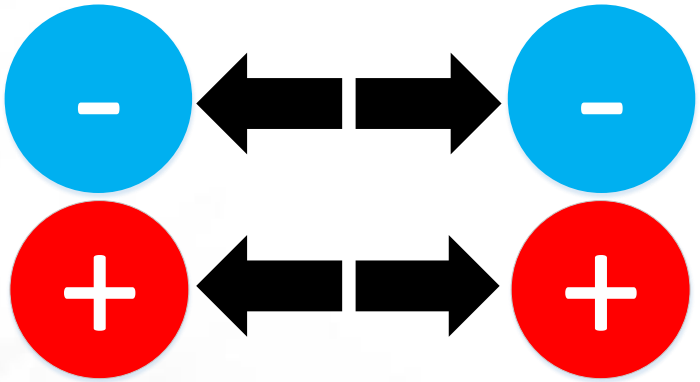
Electric Charge

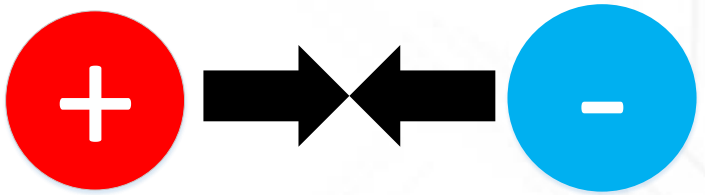
- Two types → ***Positive*** and ***Negative***



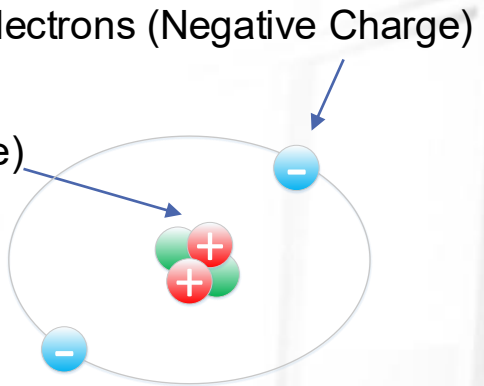


- ***Like charges repel, opposite charges attract***

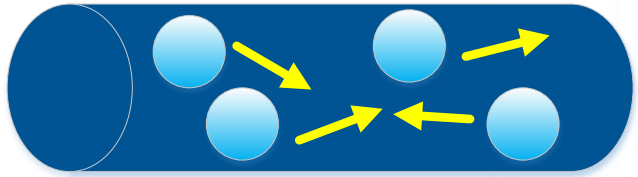




Free Electrons



- Outside forces can cause electrons to leave their orbit, causing them to become "***free electrons***"

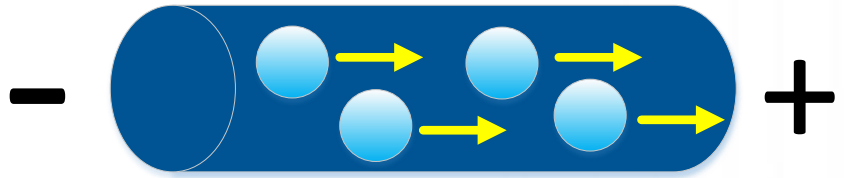


Free Electrons in Copper Wire

- When an electron leaves its orbit, the atom left behind has fewer electrons than protons giving it a positive net charge → **Positive ion**

Current

- The flow of electrons in a conductor creates ***current***



Free Electrons flowing in Copper Wire – "Current"

- **Current:** The rate of charge flow

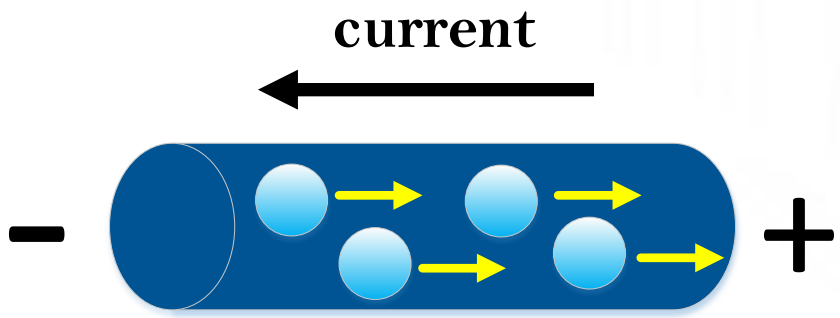
$$i = \frac{dq}{dt}$$

where i = current in **Amperes (A)**,
q = charge in Coulombs (C),
t = time in seconds (s)

Current

How do we define the **direction** of current?

Conventionally, ***current is defined to be the motion of positive charge***. In other words, opposite to the motion of electrons.

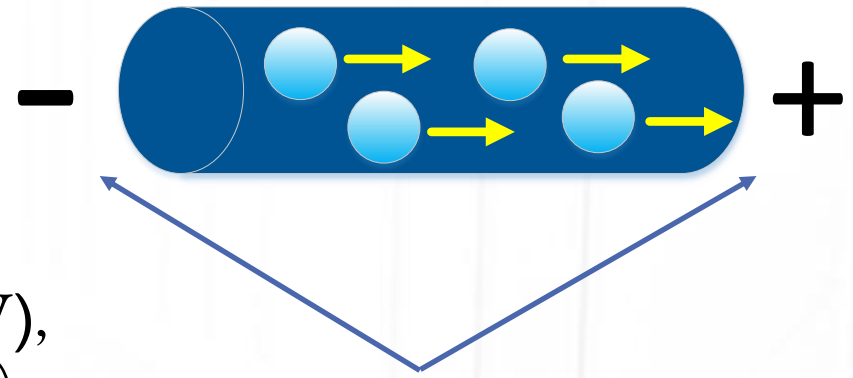


What creates current?....

Voltage

- Voltage is the electric potential energy between two points, per unit charge
- **Voltage:**

$$v = \frac{dw}{dq}$$

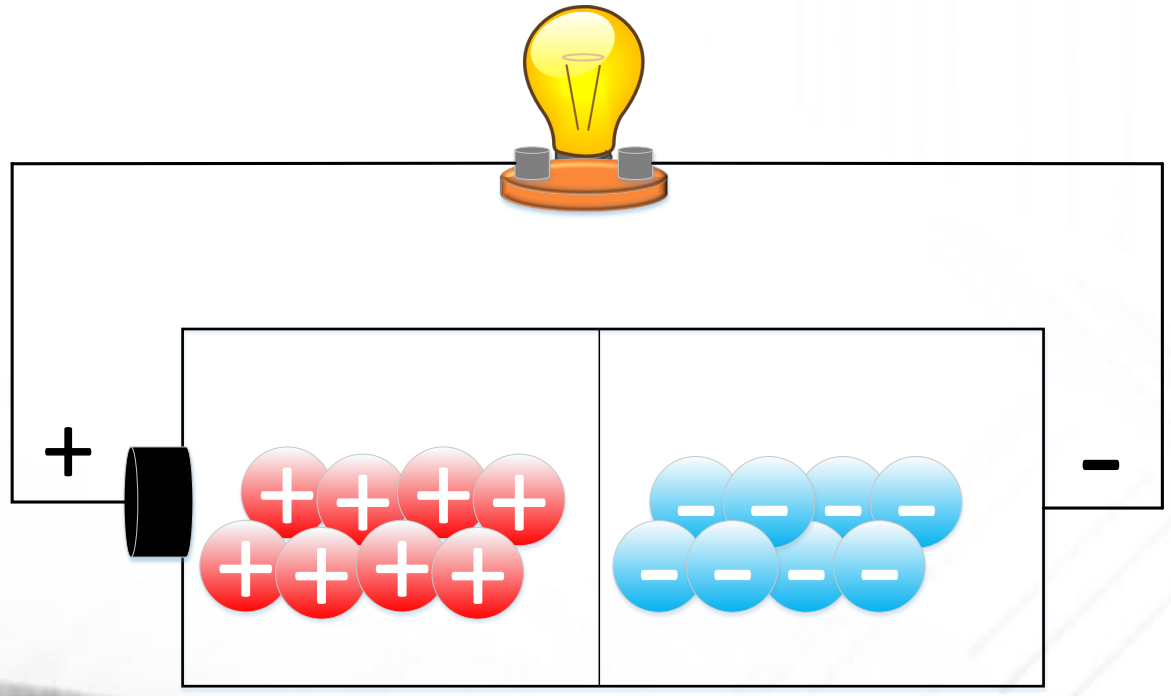


where v = voltage in **Volts (V)**,
w = energy in Joules (J),
q = charge in Coulombs (C)

Voltage provides the force to move electrons, and this is why it is also called an *electromotive force (EMF)*.

Voltage

Example: A lightbulb connected to a battery.
Current flows through the wires because the negative and positive charges want to neutralize each other. Voltage exists across the battery terminals.



Ideal Circuit Components

Ideal basic circuit elements are mathematical models of actual electrical components that have *three* properties:

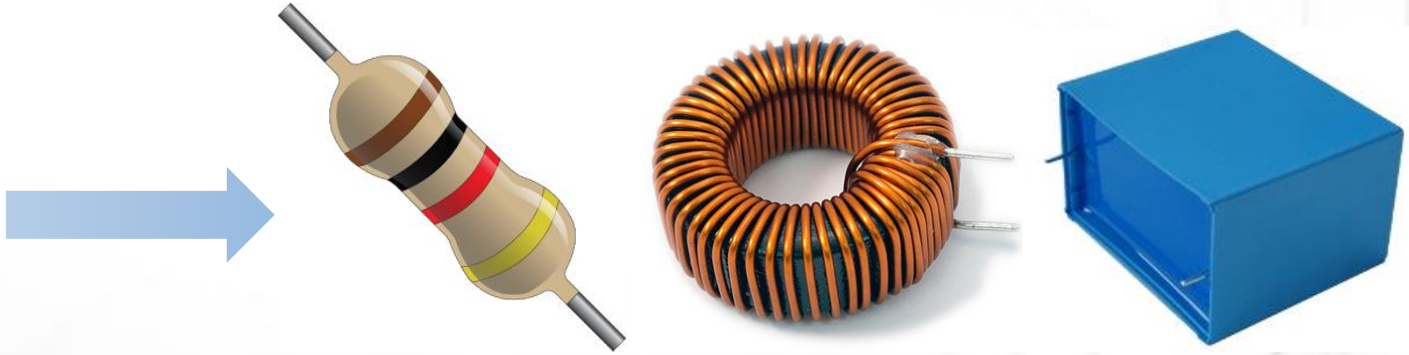
(1) They have only two terminals, which are used to connect to other circuit components

(2) They're described in terms of current and/or voltage

(3) They can't be subdivided into other elements

They must represent the behavior of the actual electrical component accurately!

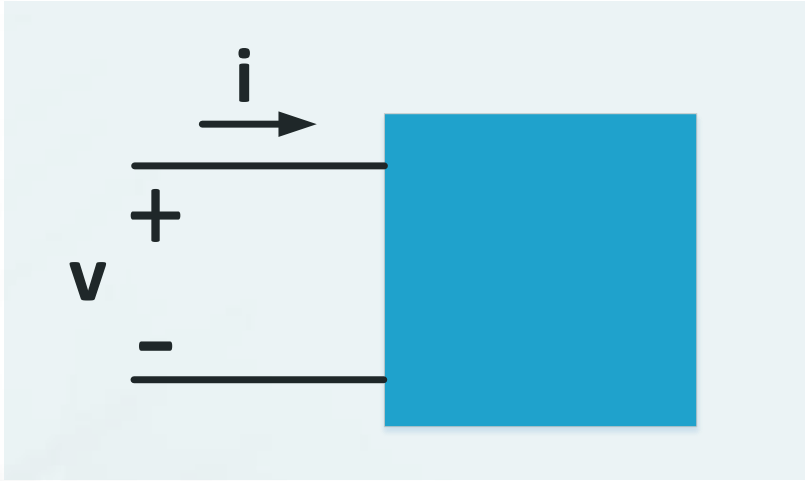
Actual Electrical Components

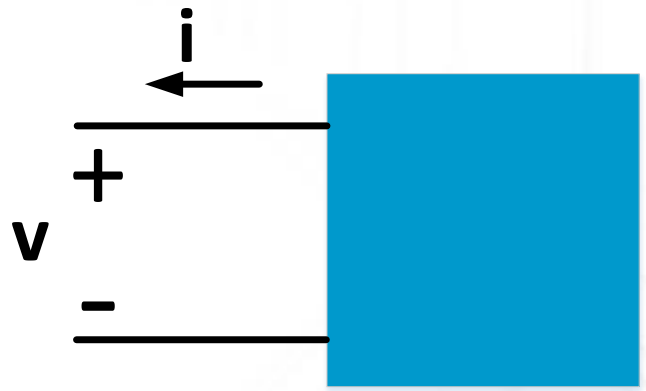


Passive Sign Convention

For elements, whenever the reference direction for the current is in the direction of the reference voltage drop across the element, use a positive sign in any equation that relates the voltage to the current.

Reference Ideal Basic Circuit Element





Power

Power is the time rate at which energy is expended or absorbed

Power:

$$\boldsymbol{p = \frac{dw}{dt}}$$

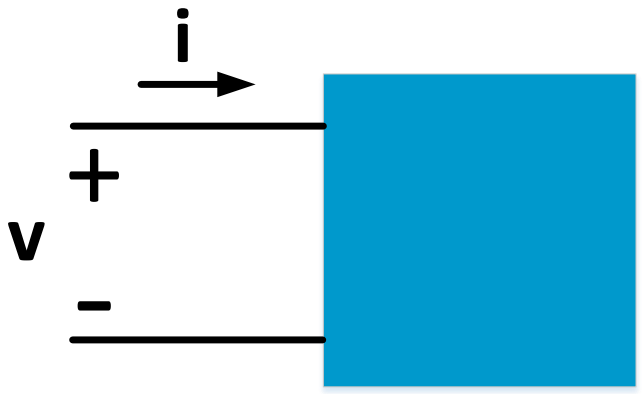
where p = power in **Watts (W)**,
w = energy in Joules (J),
t = time in seconds (s)



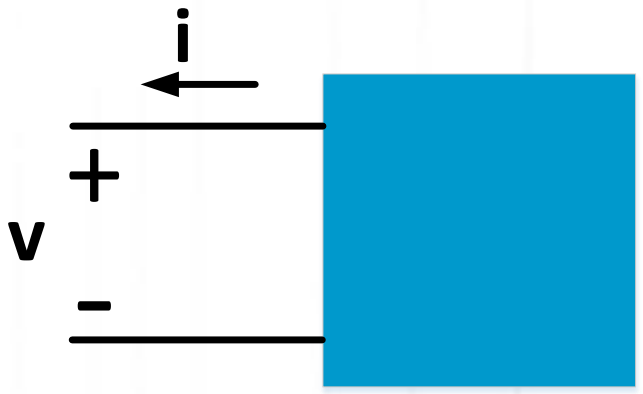
Power can be rewritten as follows:

$$\boxed{\boldsymbol{p = vi}}$$

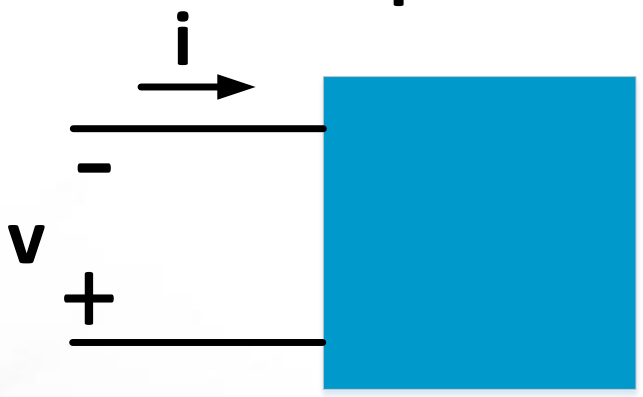
Passive Sign Convention: Part II



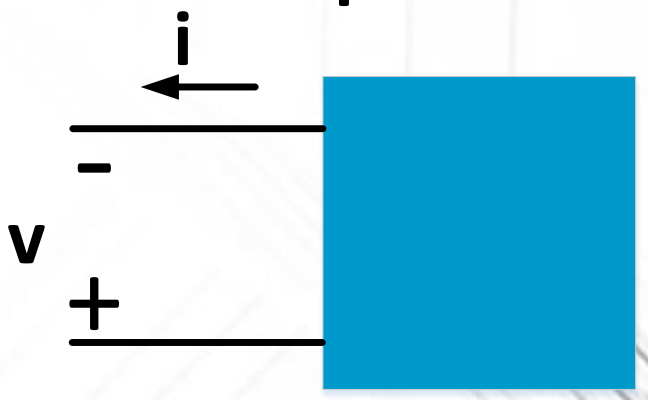
p=+vi



p=-vi



p=-vi



p=+vi

Positive/Negative Power

- If power is positive, $p > 0$, power is being ***absorbed*** by the circuit element (represented by the box)

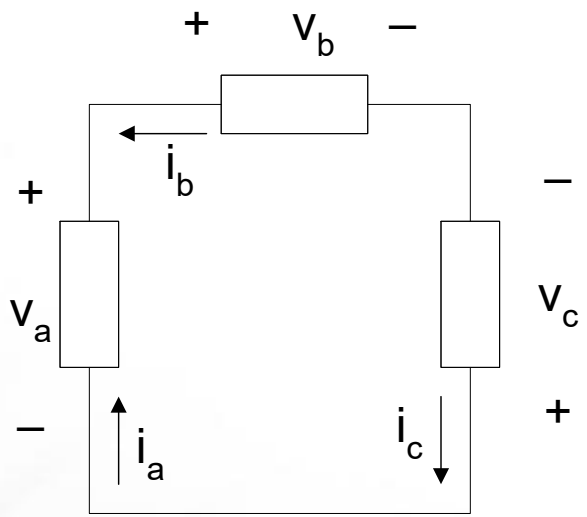
- If power is negative, $p < 0$, power is being ***generated*** by the circuit element (represented by the box)

Principle of Conservation of Energy

Total Power Delivered = Total Power Absorbed

$$\boldsymbol{\Sigma P = 0}$$

Example:



$$p_a + p_b + p_c = 0$$

Summary

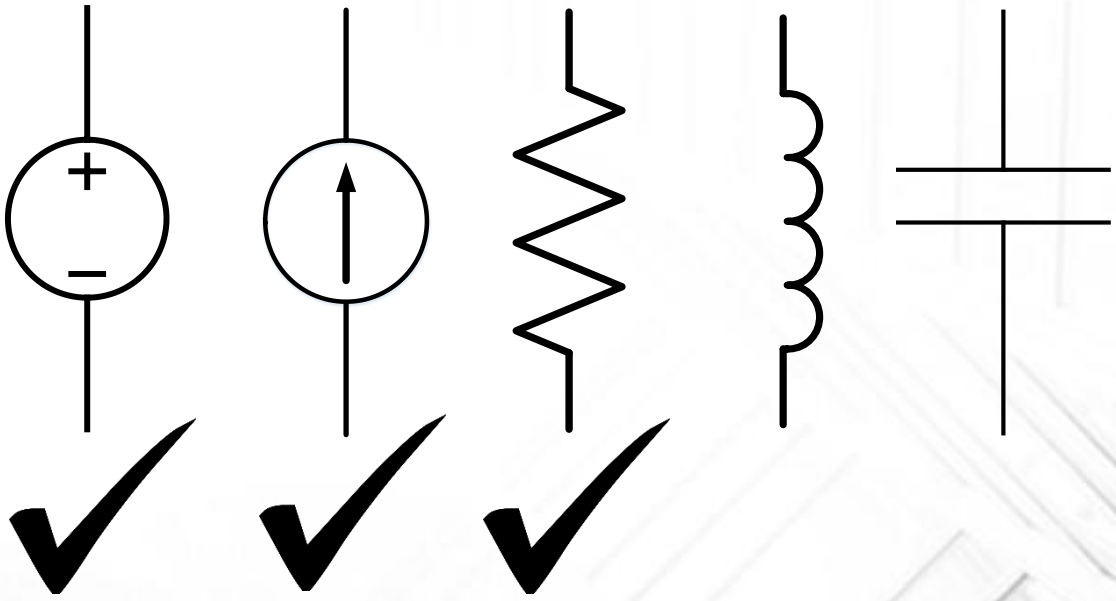
Takeaways:

- What is current? What is it's unit of measure?
- What is voltage? What is it's unit of measure?
- What is power? What is it's unit of measure? How do you interpret positive/negative power?
- What is the passive sign convention?

Ideal Basic Circuit Elements

- Which five *ideal basic circuit elements* will be covered in this course?

  - ***Voltage Sources***
  - ***Current Sources***
  - ***Resistors***
  - Inductors
  - Capacitors



Electric Source

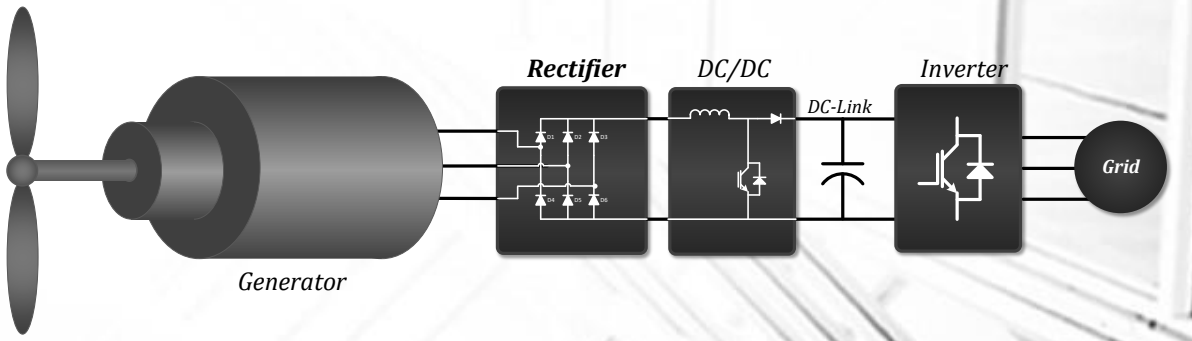
An electric source can convert nonelectric energy to electric energy and vice versa. It is a source of **either** voltage or current. In other words, it can keep either the voltage or the current constant.

Examples:

→Battery (chemical <-> electrical)



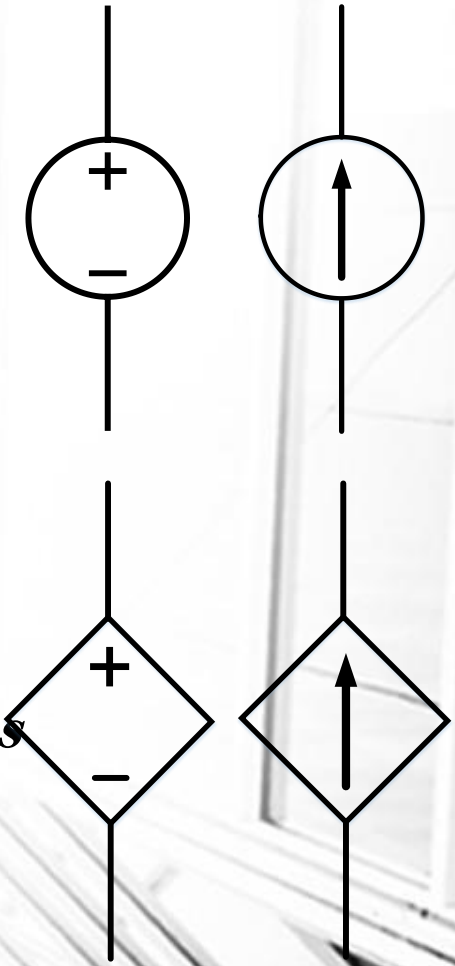
→Dynamo (mechanical <-> electrical)



Independent/Dependent Sources

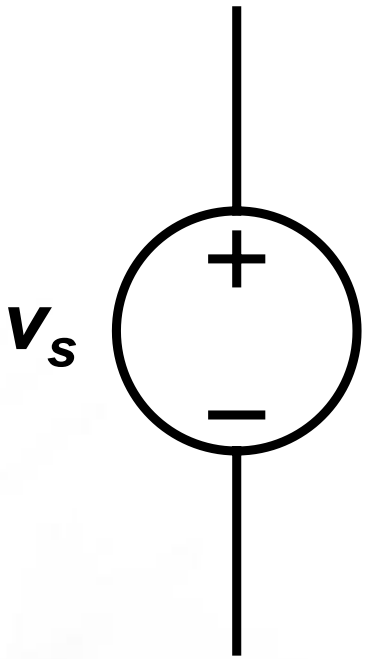
- **Independent sources** establish a voltage or current in a circuit without relying on voltages or currents elsewhere in the circuit → ***The source itself determines the value of the voltage or current that it supplies***

- **Dependent sources** establish a voltage or current whose value depends on the value of a voltage or current elsewhere in the circuit → ***Voltage or current elsewhere in the circuit determines the value of the voltage or current that it supplies***



Ideal Voltage Source

An ideal voltage source maintains a **constant voltage** across its terminals (***and has no control over the current passing through it***)

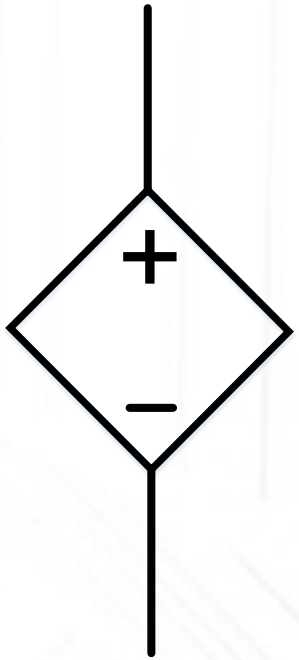


*Ideal **Independent** Voltage Source*

$v_s = \mu\, v_x$

OR

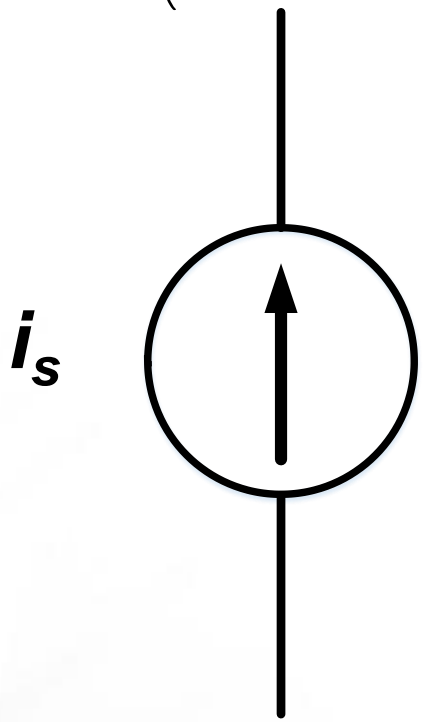
$v_s = \mu\, i_x$



*Ideal **Dependent** Voltage Source*

Ideal Current Source

An ideal current source maintains a **constant current** through its terminals (***and has no control over the voltage across it***)

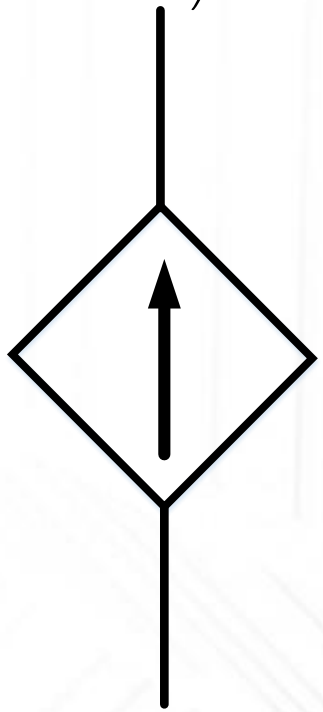


*Ideal **Independent** Current Source*

$i_s = \mu\, i_x$

OR

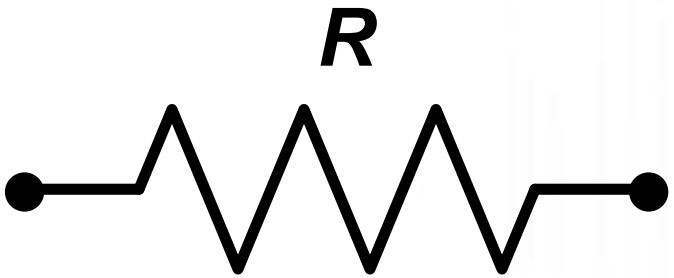
$i_s = \mu\, v_x$



*Ideal **Dependent** Current Source*

Resistor

- A **resistor** is a circuit element that exhibits ***resistance***, which means it ***impedes the flow of current***



- Unit of measurement: **Ohm (Ω)**
- Examples:



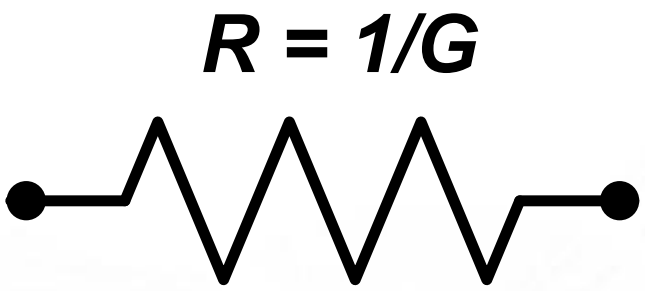


Conductance

- Conductance is the reciprocal of resistance, and represents the ease with which electric charge is conducted in a resistor

$$G = \frac{1}{R}$$

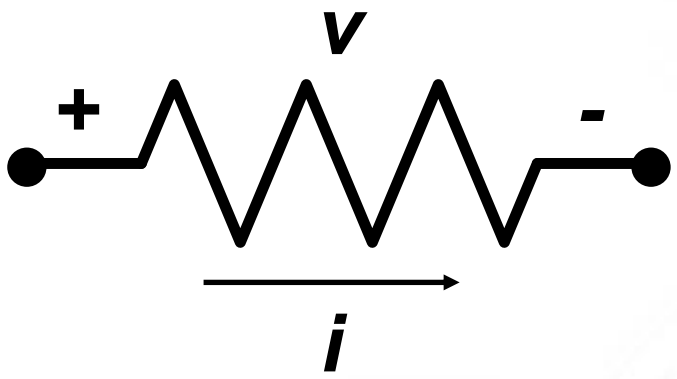
- Unit of measurement: **siemens (S) or mho (℧)**



Ohm's Law

Ohm's law defines *the relationship between the voltage across and the current through a resistor*

$$v = iR$$



Power in a Resistor

- The original equation for power was:

`

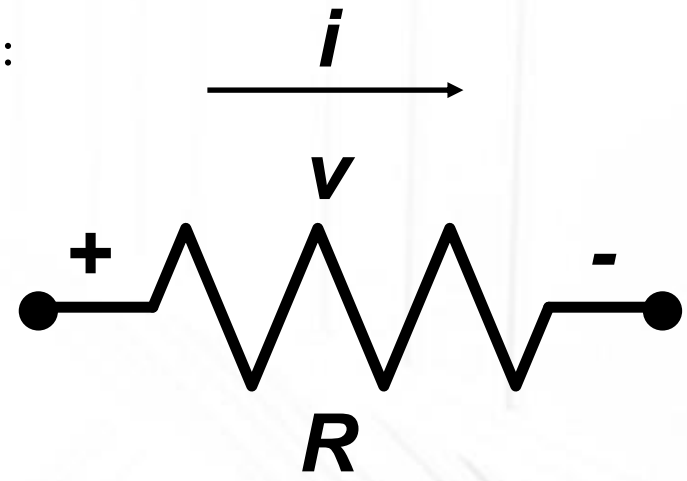
$$p = vi$$

- Now, with Ohm's law this can be rewritten as:

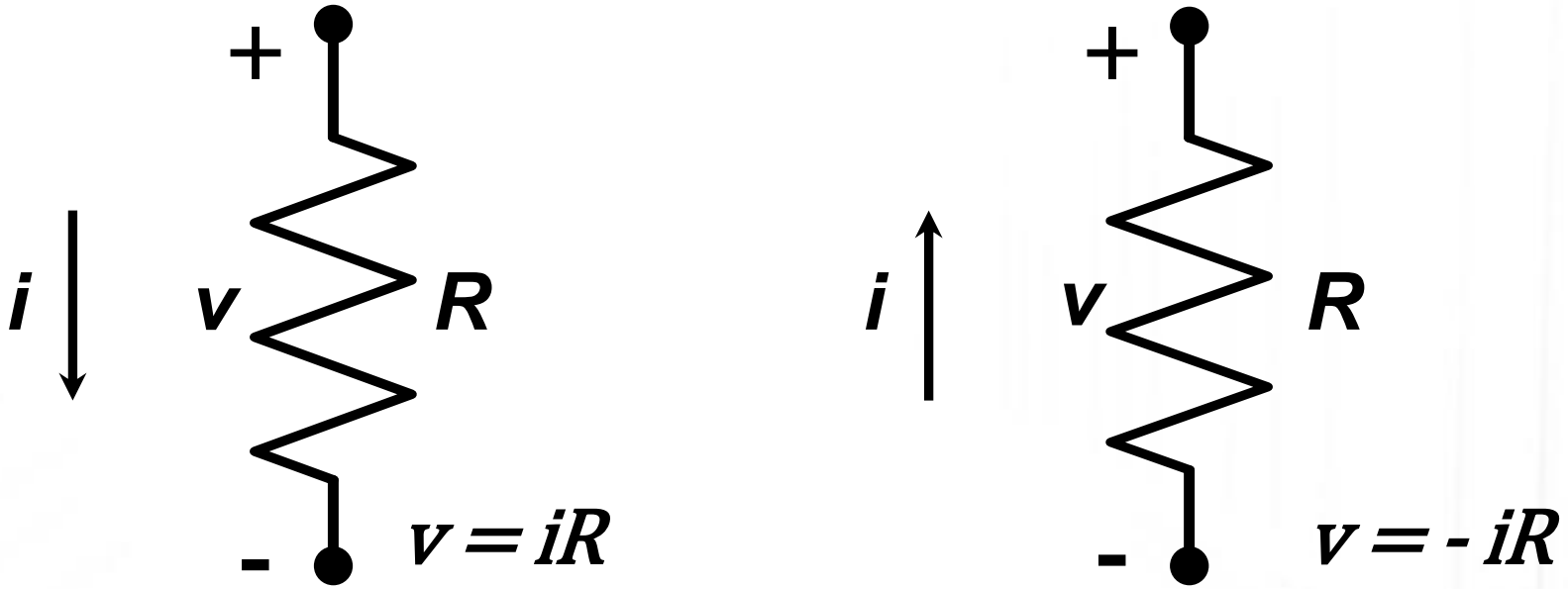
$$p = Ri^2$$

which can also be rewritten as:

$$p = \frac{v^2}{R}$$



Power: Passive Sign Convention



- Does the expression of power change in the second example? No.

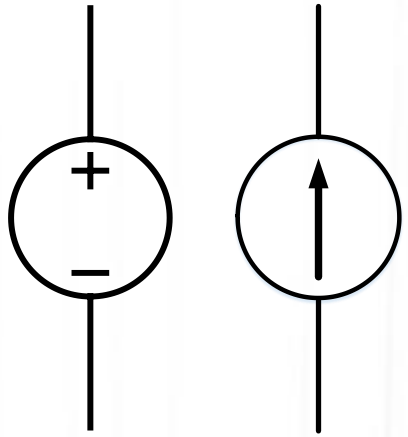
$$\boxed{p = -i(-iR) = Ri^2 = vi}$$

→Resistors don't generate power, they only absorb power.

Active and Passive Elements

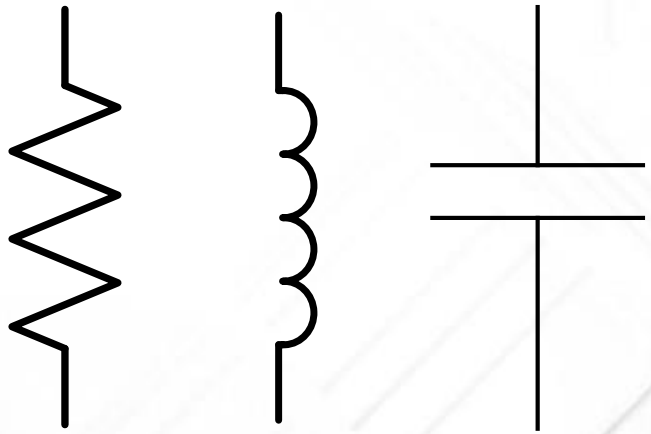
Active elements can generate electric energy.

→Voltage Sources, Current Sources



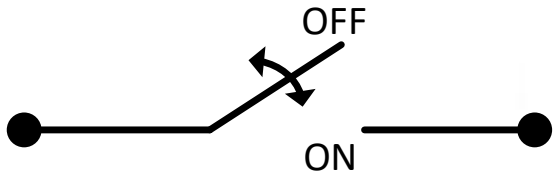
Passive elements can not generate electric energy.

→Resistors, Inductors, Capacitors



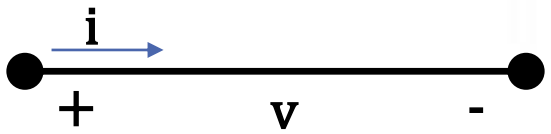
Short Circuit/Open Circuit

- What is a switch?



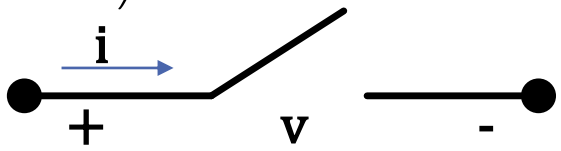
Two states: ON or OFF

Short Circuit (switch is ON): $R = 0$



$v = iR = 0$ ⟶ Voltage is Zero!

Open Circuit (switch is OFF): $R = \infty$



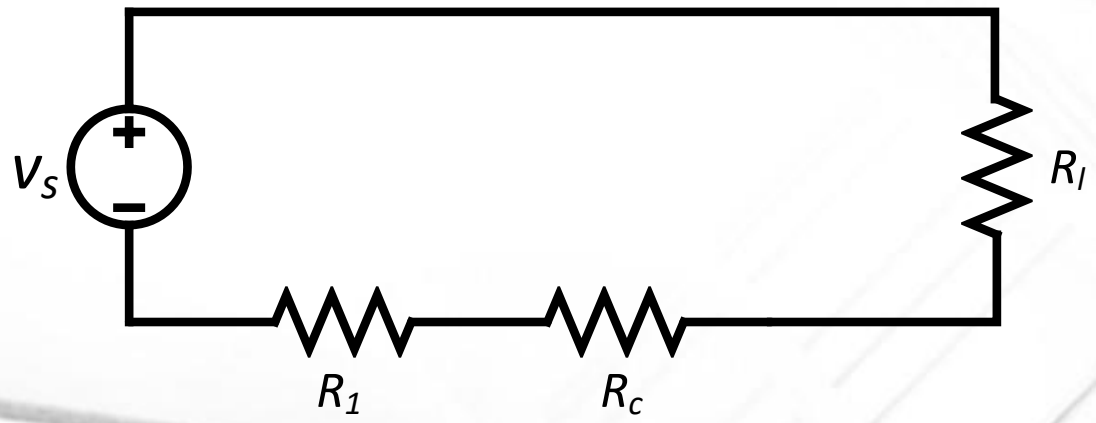
$i = {}^{v}/_{R} = 0$ ⟶ Current is Zero!

Kirchhoff's Laws

- *How do we solve a circuit?* **By determining the voltage across and the current in every element of the circuit**

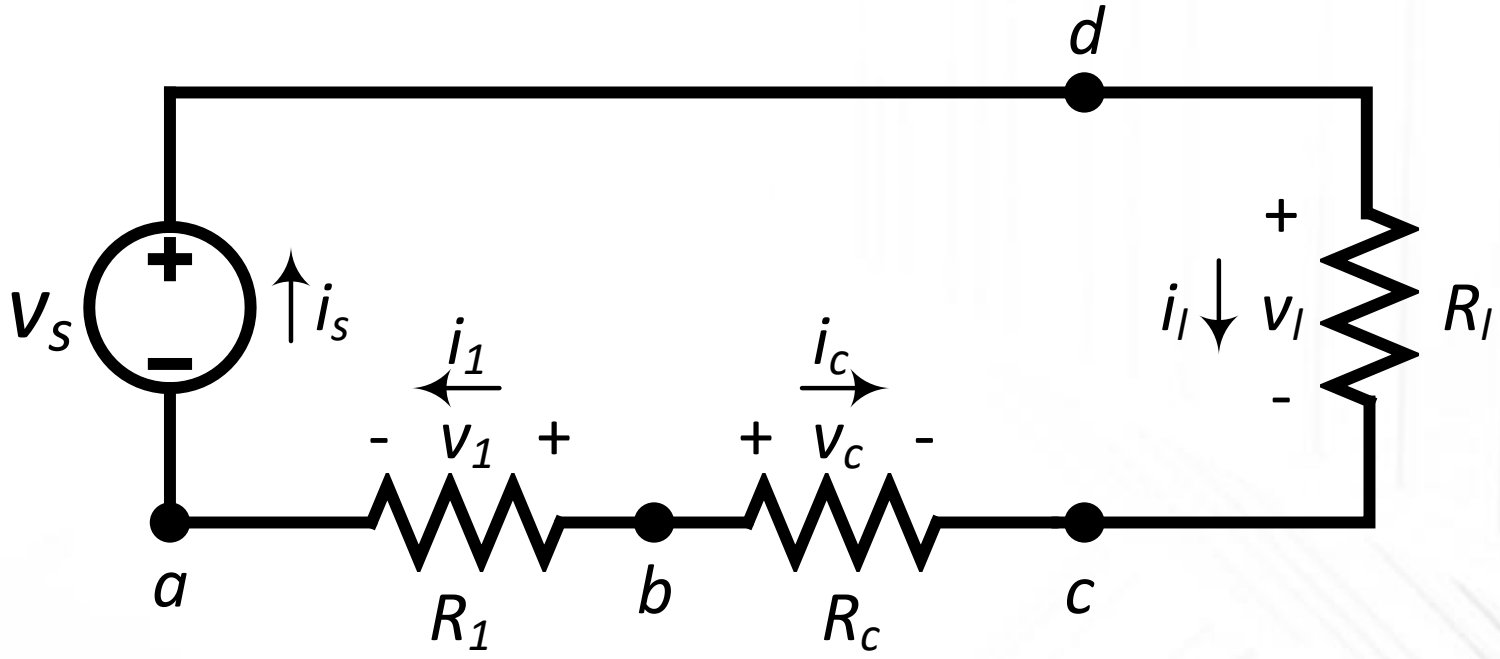
- What is a **node?** A node is *a point where two or more circuit elements meet.*

Example: Find the nodes.



Kirchhoff's Laws

Now, label the voltages (remember that you can set the voltage polarity arbitrarily), and finally the currents…



Kirchhoff's Laws

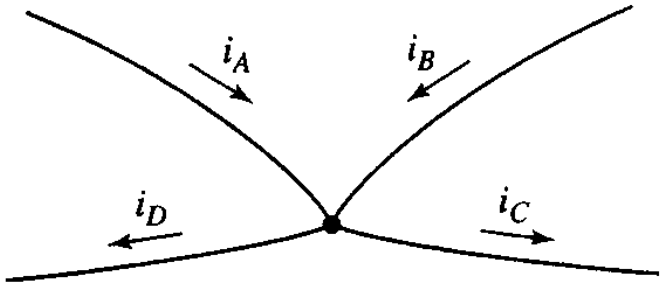
- Note that v_s, R_1, R_2, and R_3 are given in this example
- As stated earlier, **to solve the circuit the voltages across and currents in each element must be known**
- So, the following quantities are still unknown: i_s, i_1, i_c, i_l, v_1, v_c, and v_l
- We have 7 unknown variables, and need 7 equations. Ohm's law only gives us 3 equations: $v_1 = i_1 R_1$, $v_c = i_c R_c$, $v_l = i_l R_l$

…How can we find the other missing equations?

Kirchhoff's Current Law (KCL)

Kirchhoff's Current Law

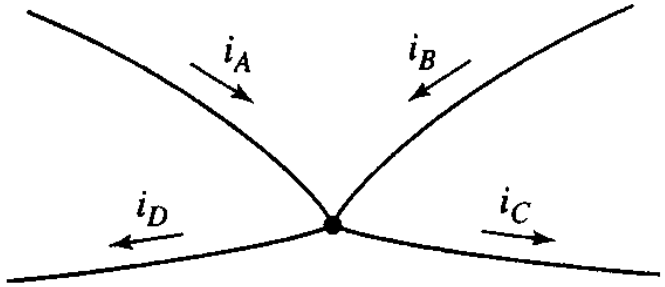
The algebraic sum of all the currents at any node in a circuit is zero.



$$i_A + i_B - i_C - i_D = 0$$

If the circuit has *n* nodes, KCL gives you *n-1* independent equations.

Kirchhoff's Current Law (KCL)



If we choose the convention that the current entering the node is negative, and the current leaving the node is positive, we get the following equation:

$$i_C + i_D - i_A - i_B = 0$$

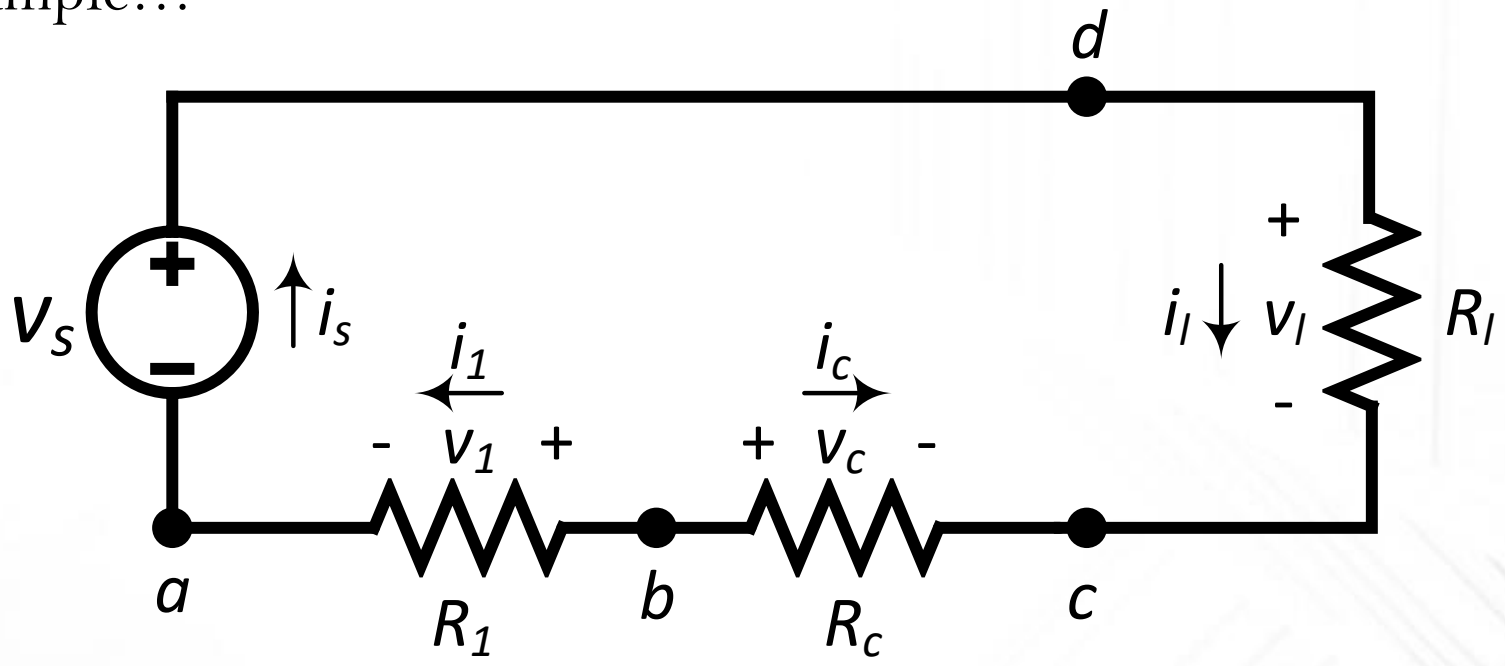
If we choose the convention that the current entering the node is positive, and the current leaving the node is negative, we get the following equation:

$$i_A + i_B - i_C - i_D = 0$$

Both are correct!

Kirchhoff's Current Law (KCL)

Let's say that we choose for the current entering the node to be negative and for the current leaving the node to be positive. Back to our example…



Kirchhoff's Current Law (KCL)

Node a: $i_s - i_1 = 0$

Node b: $i_1 + i_c = 0$

Node c: $-i_c - i_l = 0$

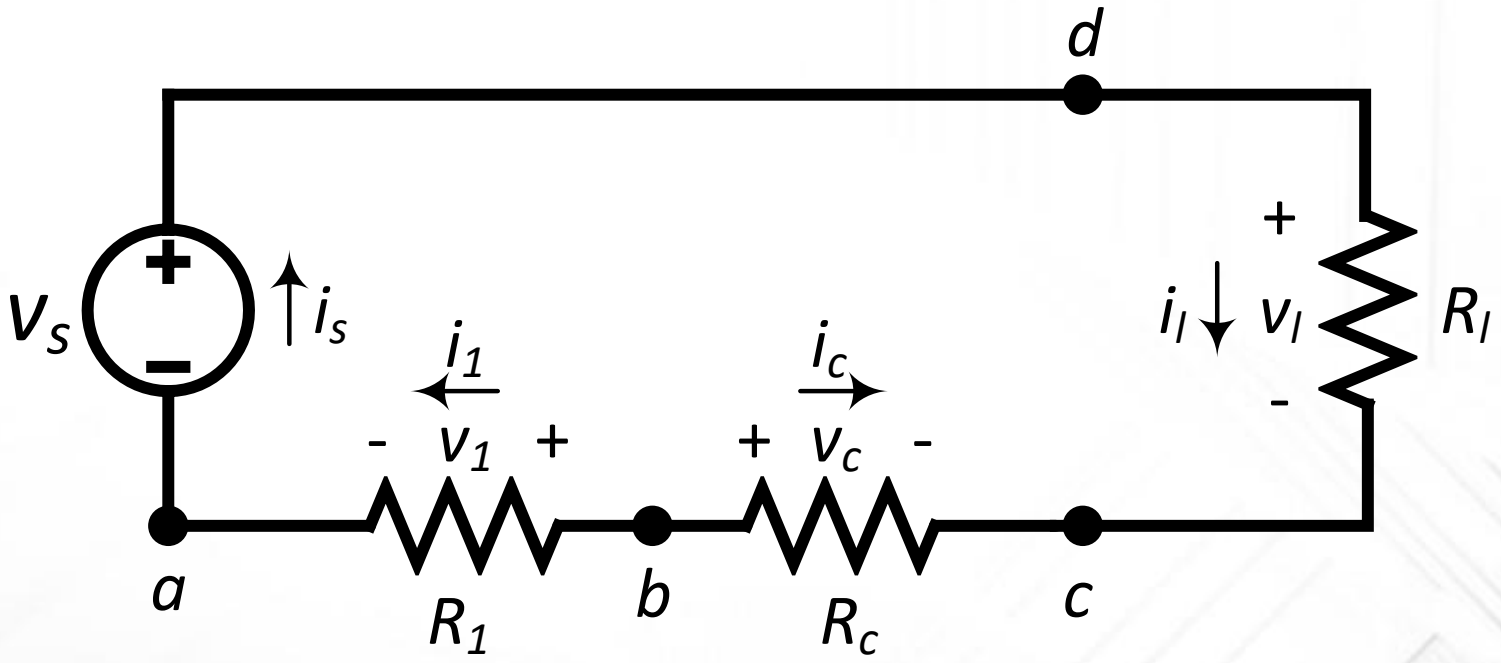
Node d: $i_l - i_s = 0$

We have 4 equations (n), but only 3 independent equations (n-1). So we need one more equation to solve the circuit…

Kirchhoff's Current Law (KCL)

In Series: When only two elements are connected to a node, these elements are said to be "in series".

Are there any elements in series in our example?



Kirchhoff's Current Law (KCL)

When two elements are in series, the current in the first element is the same as the current in the second element.

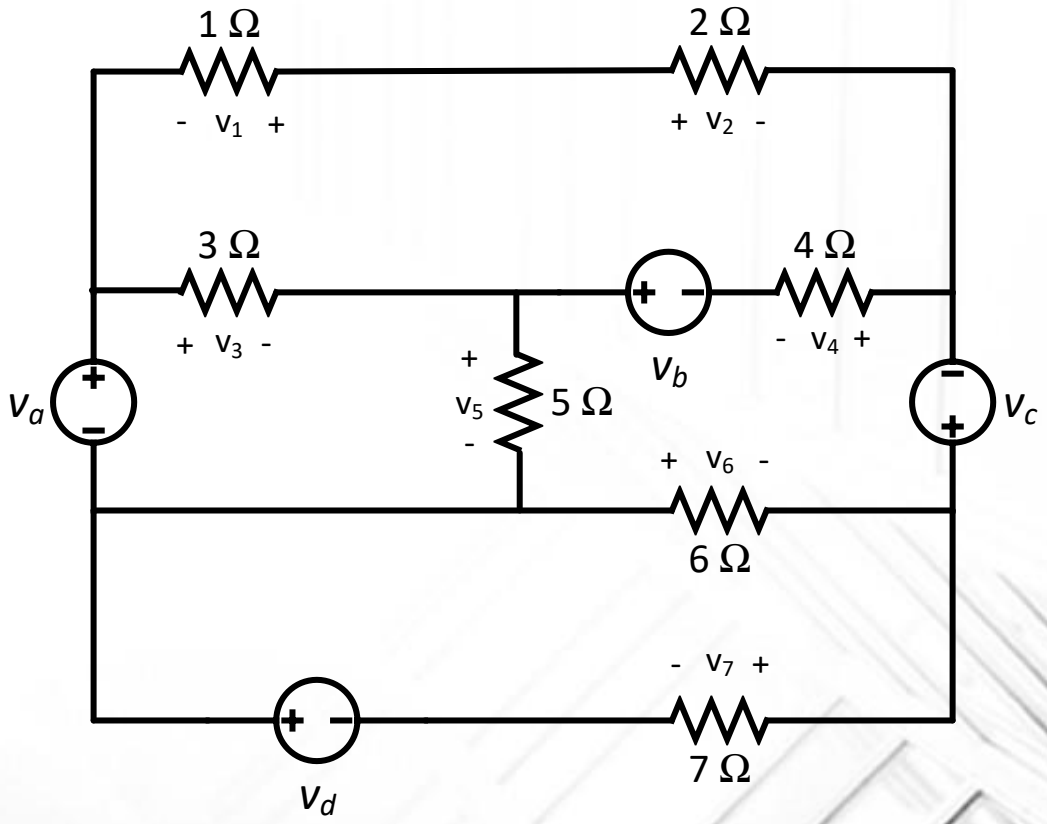
Thus,

$$i_s = i_1 = -i_c = i_l$$

Kirchhoff's Voltage Law (KVL)

Loop: A closed path in a circuit, which starts at an arbitrary node and ends at the same node without passing through any intermediate nodes more than once

Example: Identify the loops.



Kirchhoff's Voltage Law (KVL)

Kirchhoff's Voltage Law

The algebraic sum of all the voltages around any loop in a circuit is zero.

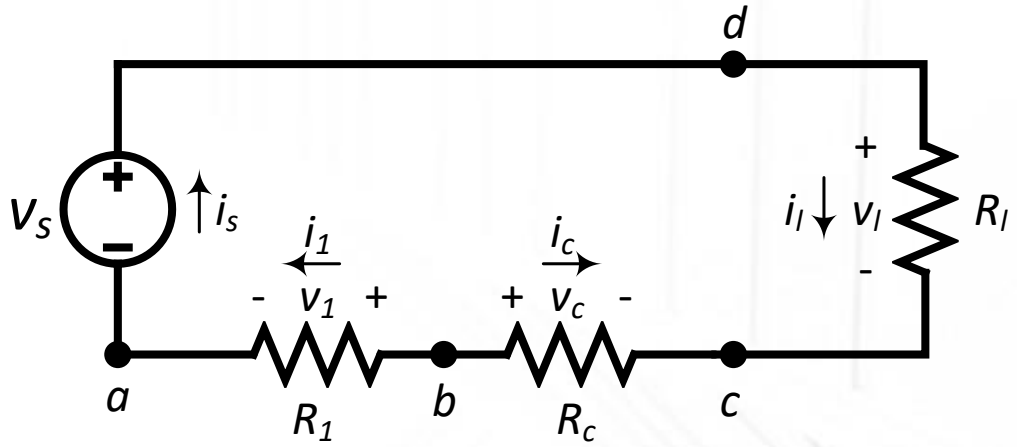
Going back to our example, what's equation(s) does KVL give us?

$$v_l - v_c + v_1 - v_s = 0$$

OR

$$-v_l + v_c - v_1 + v_s = 0$$

Both are correct!



Now, we have all 7 equations.